University of Exeter

MTH2008

Real Analysis

On Campus Exam	Closed Book
Additional Materials: None	**Permitted Materials:** Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

-The **duration** of this paper is 2 hours
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Use only black pen or pencil.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

Exam Instructions:

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline.

SECTION A

1. (a) Let S_k be defined by $S_k = (1 - \frac{1}{k}, 2 + \frac{1}{k}]$ for $k \in \mathbb{N}$.

Determine each of the following sets and express your answer as an interval or as a union of intervals:

(i) $\cup_{k=1}^{\infty} S_k$ (ii) $\cap_{k=1}^{\infty} S_k$ (iii) $\cup_{k=1}^{\infty} (S_k)^c$ (iv) $\cap_{k=1}^{\infty} (S_k)^c$.

(8)

(b) Give a counterexample to each of the following false statements. In each case explain why the example you have given is a counterexample.

(i) The isolated points of a subset of $\mathbb{R}$ form a closed set.

(ii) Every open subset of $\mathbb{R}$ contains at least two points.

(4)

(c) Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} \dfrac{|x|\sqrt{x^2 - 4x + 4}}{x - 2} & \text{if } x \neq 2 \\ 2 & \text{if } x = 2. \end{cases}$$

Show why the function f is not differentiable at $x_0 \in \{0, 2\}$, and state the set of points where f is differentiable. Justify your answers.

(10)

(d) Let $a, b \in \mathbb{R}$ with $a < b$. Consider a function $f : [a, b] \to \mathbb{R}$. Let g be defined by

$$g(x) = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0. \end{cases}$$

(i) Prove the statement: If f is continuous at $x_0 \in [a, b]$, then so is g.

(ii) Is the converse of this statement true? Justify your answer with a proof or a counterexample.

(9)

(e) For an integer k, define the set S_k by $S_k = \{k^n | n \in \mathbb{N}\}$.
Find the set K of all integers k with the property that the set S_k is bounded below. Justify your answer.

(10)

(f) Let $h : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be the function defined by

$$h(x) = \frac{x + |x|(1+x)}{x} \sin\left(\frac{1}{x}\right).$$

(i) Use the ϵ-δ definition of limits to show that $\lim_{x \to 0-} h(x) = 0$.

(ii) Show that $\lim_{x \to 0+} h(x)$ does not exist.

(9)

[**50**]

SECTION B

2. (a) Let $S = \left\{ \frac{(-1)^n}{n} \middle| n \in \mathbb{N} \right\}$.

(i) Determine the set of boundary points and the closure of the set S.

(ii) Is the set S open, closed, both or neither? Justify your answer.

(10)

(b) (i) Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions $f_n : [-1, 1] \to \mathbb{R}$ defined by

$$f_n(x) = |x|^{1+\frac{1}{n}}.$$

Prove that this sequence converges uniformly to $f(x) = |x|$ on $[-1, 1]$.

(ii) Is a uniform limit of a sequence of differentiable functions necessarily differentiable? Justify your answer with a proof or a counterexample.

(15)

[25]

3. (a) Use Riemann sums to show that the function $f : [1, 2] \to \mathbb{R}$ given by $f(x) = x^2 - 2x$ is Riemann integrable, and then evaluate $\int_1^2 f(x)dx$.

(12)

(b) Let $r \in \mathbb{R}$. Let $\{s_n\}_{n=1}^{\infty}$ be the sequence of real numbers defined by

$$s_n := \sum_{k=1}^{n} r^k.$$

(i) Use Cauchy's Convergence Criterion to prove that the sequence $\{s_n\}_{n=1}^{\infty}$ converges when $r \in (-1, 1)$.

(ii) Is this sequence bounded when $r \in (-1, 1)$? Justify your answer.

(iii) Describe the behaviour of the infinite series $\sum_{k=1}^{\infty} r^k$ when $r = 1$ and then when $r = -1$. In each case justify your answer.

(13)

[25]

4. Fix $k \in \mathbb{N}$. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by

$$f(\mathrm{v}) = (x_1^2 + x_2^2)^k,$$

where $\mathrm{v} = (x_1, x_2)^T \in \mathbb{R}^2$.

(a) Show that if $k = 2$ then the function f is continuous on $\mathbb{R}^2$. (5)

(b) Show that if $k = 2$ then the function f is differentiable at every point $\mathrm{v} \in \mathbb{R}^2$, and find its derivative. (10)

(c) Find all the stationary points of f when $k = 2$. (5)

(d) Is the function f differentiable at every point $\mathrm{v} \in \mathbb{R}^2$ when k is any natural number? Justify your answer. (5)

[**25**]

[END OF PAPER]